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BSCS 4A
Computer Vision
Assignment 6

Task 1:

The Median filter significantly outperforms Gaussian smoothing on salt-and-pepper noise because it is a nonlinear order-statistics filter, while the Gaussian filter is a linear weighted-average filter. In a 5×5 neighborhood, the Median filter sorts the 25 pixel values and selects the middle value, so extreme noise values such as 0 or 255 have little effect unless they dominate the neighborhood. In contrast, the Gaussian filter calculates a weighted average, causing these extreme values to directly influence the output and spread the noise into surrounding pixels. Therefore, the Median filter typically produces a lower MSE and higher PSNR, indicating better restoration. Qualitatively, it also preserves sharper edge boundaries, whereas Gaussian averaging blends pixels across edges and produces noticeable blurring. Thus, the Median filter is better suited for impulse noise because it removes outliers while preserving important image structures.

Task 2:

The Ideal Band-Pass filter typically produces sharper frequency separation but introduces more visible ripple or ringing around high-contrast edges, while the Butterworth Band-Pass filter provides a smoother transition between passed and rejected frequencies, resulting in fewer artifacts. The ringing in the Ideal filter is caused by its abrupt frequency cutoff, where the mask suddenly changes from 0 to 1. Mathematically, this sharp discontinuity in the frequency domain corresponds to oscillatory behavior in the spatial domain, producing the Gibbs phenomenon near sharp edges. In contrast, the Butterworth filter has a gradual frequency transition controlled by its order, reducing the abrupt discontinuity and therefore minimizing ringing. Consequently, the Ideal filter preserves a sharper but potentially more oscillatory response, whereas the Butterworth filter produces a smoother spatial reconstruction with fewer ripple artifacts.

Task 3:

When the high-boost factor k is significantly higher than 2.0, unsharp masking produces very strong sharpening, making edges appear more pronounced but also creating bright/dark halos, overshoot, and amplified noise, which can make the image look unnatural. In a flat homogeneous region, the original and blurred pixels are very similar, so the unsharp mask $M = I - G * I$ is close to zero; therefore, the pixel standard deviation increases only slightly. In high-contrast boundary regions, however, the difference between the original and blurred image is much larger, and multiplying this difference by a large k greatly amplifies local intensity variations. Consequently, the standard deviation increases substantially near edges, reflecting stronger contrast and sharper boundaries, while excessive k can cause clipping at 0 and 255 and produce visible artifacts.